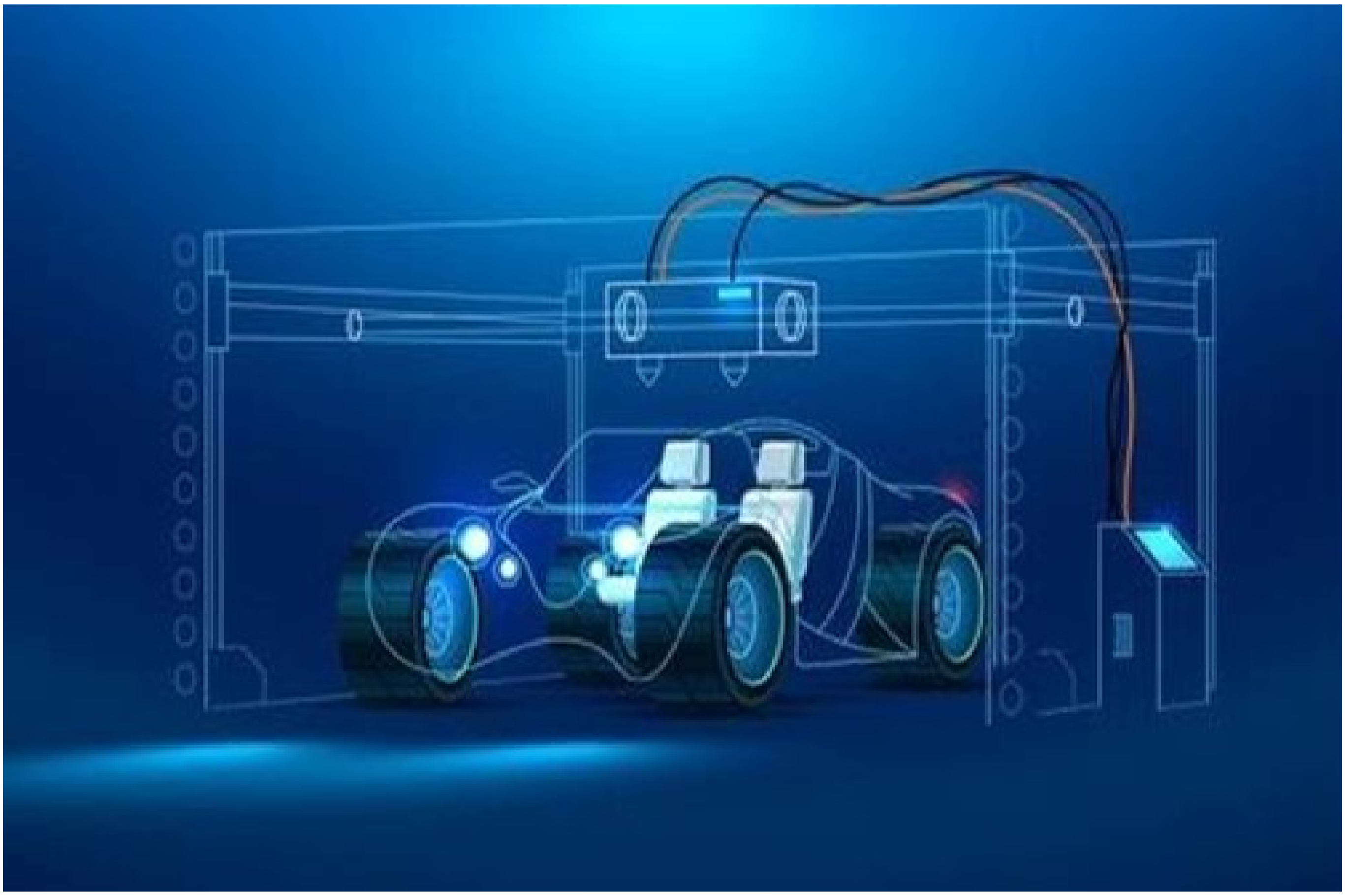
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

Qualification Pack



Additive Manufacturing

QP Code: ASC/Q6420

Version: 1.0

NSQF Level: 4.5

Automotive Skills Development Council || 153, GF, Okhla Industrial Area, Phase 3
New Delhi 110020 || email:ayushi@asdc.org.in

Contents

ASC/Q6420: Additive Manufacturing

Brief Job Description

The individual in this job is responsible for preparing object model of the part/product and manufacturing of parts/product on 3D printing machine.

Personal Attributes

The individual on the job needs to have factual knowledge of: Product designing processes. Different types of software used in the heat treatment process and their identification. Quality check of product design. 3D printing of automotive parts

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ASC/N9821: Work organization and management (Additive Manufacturing)
2. ASC/N9822: Communication and interpersonal skills (Additive Manufacturing)
3. ASC/N6457: Three D digitizing
4. ASC/N6453: Component optimization/structural optimization
5. ASC/N6454: Transfer-to-CAD and optimization
6. ASC/N6455: Preparation and forming
7. ASC/N6456: Finalize and deliver workpieces

Qualification Pack (QP) Parameters

Sector	Automotive
Sub-Sector	Manufacturing and R&D
Occupation	Production Engineering
Country	India
NSQF Level	4.5
Credits	17

Aligned to NCO/ISCO/ISIC Code	NCO-2015/2144.0801
Minimum Educational Qualification & Experience	No formal education prescribed with of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	25 Years
Last Reviewed On	NA
Next Review Date	08/02/2026
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-AU-01822-2024-V1-ASDC
NQR Version	1

ASC/N9821: Work organization and management (Additive Manufacturing)

Description

This NOS unit is about implementing safety, planning work, adopting sustainable practices for optimising the use of resources.

Scope

The scope covers the following :

- Manage work and resources

Elements and Performance Criteria

Manage work and resources

To be competent, the user/individual on the job must be able to:

PC1. Provide and maintain a safe, tidy, and efficient work area
PC2. Promote health and safety legislation, best working practice, and environmental protection
PC3. Apply the internationally recognized standards (ISO) and standards currently used and recognized by industry
PC4. Use planning and time management during the work
PC5. Prioritize between work demands on a rational basis
PC6. Independently Interpret technical tasks
PC7. Estimate and plan the time, sequence, and duration of tasks and steps
PC8. Produce work that fully meets the technical specifications and standards
PC9. Create and apply innovative and creative solutions to problems and challenges of AM
PC10. Maintain a productive appearance and demeanor
PC11. Work efficiently, economically, and data rationally

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Standards for environmental protection, safety, hygiene and accident prevention related to the use of, optical and laser 3D scanners, graphic work stations, additive machines, other machines, and post-processing equipment
KU2. The principles and applications of additive manufacturing (AM)
KU3. The principles and applications of related and replacement technologies
KU4. The importance of planning and time management during work
KU5. The importance of prioritizing
KU6. The importance of cost accounting and analysis

KU7. Current internationally recognized standards (ISO), and standards currently used and recognized by industry

KU8. The role and significance of providing innovative and creative solutions to technical and design problems and challenges

KU9. The importance of maintaining a productive and professional demeanour

KU10. The importance of efficient, economical, and data rational work habits and performance

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Manage work and resources	**15**	**25**	**-**	**10**
PC1. Provide and maintain a safe, tidy, and efficient work area	2	3	-	1
PC2. Promote health and safety legislation, best working practice, and environmental protection	1	2	-	1
PC3. Apply the internationally recognized standards (ISO) and standards currently used and recognized by industry	2	3	-	1
PC4. Use planning and time management during the work	1	2	-	1
PC5. Prioritize between work demands on a rational basis	1	2	-	1
PC6. Independently Interpret technical tasks	1	2	-	1
PC7. Estimate and plan the time, sequence, and duration of tasks and steps	2	2	-	1
PC8. Produce work that fully meets the technical specifications and standards	2	3	-	1
PC9. Create and apply innovative and creative solutions to problems and challenges of AM	1	2	-	1
PC10. Maintain a productive appearance and demeanor	1	2	-	1
PC11. Work efficiently, economically, and data rationally	1	2	-	-
NOS Total	**15**	**25**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9821
NOS Name	Work organization and management (Additive Manufacturing)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N9822: Communication and interpersonal skills (Additive Manufacturing)

Description

This unit is about communicating with team members, superior and others.

Scope

The scope covers the following :

- Communicate effectively with others

Elements and Performance Criteria

Communicate effectively with others

To be competent, the user/individual on the job must be able to:

PC1. Communicate effectively, using strong inter-personal skills with coworkers, clients, and other related professionals to ensure that developing projects meet requirements

PC2. Read, interpret, and extract technical data and instructions from any available sources

PC3. Use discretion and confidentiality when dealing with clients

PC4. Clarify terms of reference, specifications, and instructions, for the most accurate implementation of requirements

PC5. Maintain proactive continuous professional development in order to sustain knowledge and skill in new and developing technologies and practices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The importance of effective communication and inter-personal skills between co-workers, customers, and other related professionals

KU2. The range of purposes of documentation in both paper and electronic forms as well as instructions in any forms

KU3. Technical terminology and symbols

KU4. The importance of technical specifications

KU5. The importance of resolving misunderstandings and conflicting demands

KU6. The importance of gaining, retaining, and developing knowledge through log books, exhibitions, articles, and specialist internet resources

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Communicate effectively with others	**20**	**30**	**-**	**10**
PC1. Communicate effectively, using strong inter-personal skills with coworkers, clients, and other related professionals to ensure that developing projects meet requirements	3	5	-	2
PC2. Read, interpret, and extract technical data and instructions from any available sources	3	5	-	2
PC3. Use discretion and confidentiality when dealing with clients	3	5	-	2
PC4. Clarify terms of reference, specifications, and instructions, for the most accurate implementation of requirements	5	5	-	2
PC5. Maintain proactive continuous professional development in order to sustain knowledge and skill in new and developing technologies and practices	5	10	-	2
NOS Total	**20**	**30**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N9822
NOS Name	Communication and interpersonal skills (Additive Manufacturing)
Sector	Automotive
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N6457: Three D digitizing

Description

This NOS is about making decisions for 3D digitizing and and preparing for additive manufacturing process.

Scope

The scope covers the following :

- Carry out 3D digitizing
- Preparing measuring instruments

Elements and Performance Criteria

Carry out 3D digitizing

To be competent, the user/individual on the job must be able to:

PC1. Perform equipment's adjustment and calibration

PC2. Make decisions regarding the possibility of optical 3D digitizing, for technical reasons: Possible or impossible to perform, The accuracy it is possible to attain for the object, The required conditions for digitizing

PC3. Make decisions regarding pre-process works such as disassembly, washing, and painting

PC4. Perform pre-process actions for applying matting coating

PC5. Apply matting coating

PC6. Apply optical marks

PC7. Fix objects for subsequent digitization

PC8. Perform 3D digitizing for various objects with: different materials, different surface characteristics, different geometrical complexity

PC9. Save results in the required form

Preparing measuring instruments

To be competent, the user/individual on the job must be able to:

PC10. Prepare objects and measuring instruments for measurements

PC11. Calibrate, adjust, and align measuring instruments

PC12. Select correct measuring instruments and devices (styluses, probes, etc.), auxiliary and fixing devices (vices, V-blocks, clamps, etc.), relating to measurement strategy

PC13. Perform measurements using various control and measuring instruments

PC14. Read the indications of measuring instruments

PC15. Identify and estimate the correctness of measurements and the reliability of the data obtained, to minimize associated human factor errors

PC16. Find the required information in specialized reference books, tables, or diagrams

PC17. Carry out routine maintenance of measuring instruments

PC18. Transfer measuring data to CAD-models

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The principles of equipment operation for 3D digitizing

KU2. The advantages and disadvantages of various types of equipment for 3D digitizing, and the technologies on which they are based

KU3. The requirements for ensuring the feasibility of work and its required quality and accuracy relative to dust, base vibration, stray light sources, mobility of objects

KU4. The importance of calibration of equipment, and the requirements for calibration and digitizing conditions

KU5. The requirements of each items surface characteristics for optical 3D digitizing, relative to its purposes and uses such as fit, smoothness, transparency, translucence, glossiness

KU6. Methods and techniques for surface preparation for optical 3D digitizing, such as washing, sandblasting, and matting

KU7. The types of rejection of 3D digitizing, their sources, and ways to eliminate them

KU8. The types and range of measuring instruments and devices (probes, sensors, fixing devices, etc.)

KU9. Constructive and metrological characteristics of measuring instruments, including special ones (for measuring narrow grooves, gears, threads, etc.)

KU10. Factors influencing the reliability of the results of measurement (surface contamination, temperature imbalance, incorrect measuring force, etc.)

KU11. Measurements making methods

KU12. How to use specialized reference books, tables, or diagrams

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Carry out 3D digitizing	**16**	**29**	**-**	**11**
PC1. Perform equipment's adjustment and calibration	1	2	-	1
PC2. Make decisions regarding the possibility of optical 3D digitizing, for technical reasons: Possible or impossible to perform, The accuracy it is possible to attain for the object, The required conditions for digitizing	2	3	-	2
PC3. Make decisions regarding pre-process works such as disassembly, washing, and painting	1	2	-	1
PC4. Perform pre-process actions for applying matting coating	2	4	-	1
PC5. Apply matting coating	2	4	-	1
PC6. Apply optical marks	2	4	-	1
PC7. Fix objects for subsequent digitization	2	4	-	1
PC8. Perform 3D digitizing for various objects with: different materials, different surface characteristics, different geometrical complexity	2	4	-	2
PC9. Save results in the required form	2	2	-	1
Preparing measuring instruments	**14**	**21**	**-**	**9**
PC10. Prepare objects and measuring instruments for measurements	2	3	-	1
PC11. Calibrate, adjust, and align measuring instruments	2	3	-	1
PC12. Select correct measuring instruments and devices (styluses, probes, etc.), auxiliary and fixing devices (vices, V-blocks, clamps, etc.), relating to measurement strategy	2	3	-	1
PC13. Perform measurements using various control and measuring instruments	2	4	-	1

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Read the indications of measuring instruments	1	1	-	1
PC15. Identify and estimate the correctness of measurements and the reliability of the data obtained, to minimize associated human factor errors	2	3	-	1
PC16. Find the required information in specialized reference books, tables, or diagrams	1	1	-	1
PC17. Carry out routine maintenance of measuring instruments	1	2	-	1
PC18. Transfer measuring data to CAD-models	1	1	-	1
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6457
NOS Name	Three D digitizing
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N6453: Component optimization/structural optimization

Description

This NOS unit is about optimizing the components/structure of the design.

Scope

The scope covers the following :

- Component optimization/structural optimization

Elements and Performance Criteria

Component optimization/structural optimization

To be competent, the user/individual on the job must be able to:

PC1. Select the right type of optimization for the task in hand

PC2. Define and apply correct boundary conditions

PC3. Carry out component optimizations by applying the given optimization objectives

PC4. Evaluate the results of the optimizations with reference to quality and compliance with the given input variables

PC5. Convert the optimized components into printable components with adapted geometry

PC6. Design components according to given manufacturing processes and exploit the potential of the process for design

PC7. Optimise designs with regard to the required number of components

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Types and methods are available for software-supported component optimization

KU2. How the methods differ from each other

KU3. The results that the various methods deliver

KU4. The input variables that must be known for the procedures

KU5. The objectives pursued for structural optimization and the ways in which the results differ for the different optimizations

KU6. How optimization methods are selected and used

KU7. The valid and common rules to be followed for optimization in additive manufacturing

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Component optimization/structural optimization	**20**	**30**	**-**	**10**
PC1. Select the right type of optimization for the task in hand	2	3	-	1
PC2. Define and apply correct boundary conditions	2	3	-	1
PC3. Carry out component optimizations by applying the given optimization objectives	2	4	-	2
PC4. Evaluate the results of the optimizations with reference to quality and compliance with the given input variables	2	4	-	2
PC5. Convert the optimized components into printable components with adapted geometry	2	5	-	1
PC6. Design components according to given manufacturing processes and exploit the potential of the process for design	5	5	-	2
PC7. Optimise designs with regard to the required number of components	5	5	-	1
NOS Total	**20**	**30**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6453
NOS Name	Component optimization/structural optimization
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N6454: Transfer-to-CAD and optimization

Description

This NOS is about creating and optimizing CAD models

Scope

The scope covers the following :

- Creating and optimizing CAD models

Elements and Performance Criteria

Creating and optimizing CAD models

To be competent, the user/individual on the job must be able to:

PC1. Create editable CAD models by digitized data (polygonal models)
PC2. Apply mathematics to additive technologies
PC3. Restore missing data of the elements of objects to be redesigned, from available data: Of polygonal models (for example, the gear wheel has only one preserved tooth, the worm has only one turn, or there is only one-third of the flange), Taken from connected parts, Taken from existing objects by manual measuring (for example, the depth of a blind hole)
PC4. Change the geometry of created models according to task
PC5. Consider the features of AM and subsequent finishing processing
PC6. Analyse and optimize the structure of the model in accordance with the terms of reference
PC7. Analyse the deviation of created models from the results of 3D scanning
PC8. Provide lattices' and surfaces' topology, with analysis and optimization according to task
PC9. Apply standards for conventional dimensioning and tolerances, and geometric dimensioning and tolerance appropriate to the ISO standard
PC10. Distinguish between, and mix, polygon meshing and standard brep functionalities in a smart way

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The purposes of Transfer-to-CAD processes in relation to additive technologies (the number of parts reducing, weight reducing, functions optimizing, etc.)
KU2. Transfer-to-CAD software applications i.e. CAD, CAE, and optimization software
KU3. Mathematics, especially geometry related to additive technologies
KU4. Polygonal models requirements for Transfer-to-CAD purposes
KU5. Methods of extracting primitives from polygonal models for the purpose of restoring CAD models and their optimization
KU6. Mechanical systems and operational principles

KU7. Fundamentals of technical drafts and drawings

KU8. The basics of component assembly

KU9. Comparative methods for CAD and polygonal modelling

KU10. The requirements for CAD models for AM purposes, postprocessing, and subsequent use

KU11. AM and mechanical engineering material properties

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Creating and optimizing CAD models	**30**	**50**	**-**	**20**
PC1. Create editable CAD models by digitized data (polygonal models)	3	5	-	2
PC2. Apply mathematics to additive technologies	3	5	-	2
PC3. Restore missing data of the elements of objects to be redesigned, from available data: Of polygonal models (for example, the gear wheel has only one preserved tooth, the worm has only one turn, or there is only one-third of the flange), Taken from connected parts, Taken from existing objects by manual measuring (for example, the depth of a blind hole)	3	5	-	2
PC4. Change the geometry of created models according to task	3	5	-	2
PC5. Consider the features of AM and subsequent finishing processing	3	5	-	2
PC6. Analyse and optimize the structure of the model in accordance with the terms of reference	3	5	-	2
PC7. Analyse the deviation of created models from the results of 3D scanning	3	5	-	2
PC8. Provide lattices' and surfaces' topology, with analysis and optimization according to task	3	5	-	2
PC9. Apply standards for conventional dimensioning and tolerances, and geometric dimensioning and tolerance appropriate to the ISO standard	3	5	-	2
PC10. Distinguish between, and mix, polygon meshing and standard brep functionalities in a smart way	3	5	-	2
NOS Total	**30**	**50**	**-**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6454
NOS Name	Transfer-to-CAD and optimization
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N6455: Preparation and forming

Description

This NOS unit is about preparing and forming the system.

Scope

The scope covers the following :

- Preparing and forming the system

Elements and Performance Criteria

Preparing and forming the system

To be competent, the user/individual on the job must be able to:

PC1. investigate whether the robot and its peripheral equipment are responding to the programs' instructions

PC2. revise, repair or expand existing programs to increase operational efficiency or adapt to new requirements

PC3. repair or replace components as required

PC4. develop Human-Machine-Interface (HMI) applications for the users of the robot system, using HTML or other web technologies

PC5. advise on maintenance regimes to maximize efficiency and minimize disruption

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Physics and chemistry related to additive technologies

KU2. Model preparation, simulation and analysis software

KU3. The advantages and disadvantages of the most common additive technologies (SLS, SLM, SLA/DLP, FDM/FFF and MJ)

KU4. Properties, advantages and disadvantages of industrial materials for 3D printing

KU5. Requirements for models in accordance with technologies and materials

KU6. Post-processing technologies, their capabilities and requirements for built models (fastening requirements, binding elements, postprocessing allowances, stresses relieve operations sequence)

KU7. Technologies and processes that can used for AM parts (casting in SPF, burned or lost wax models casting, polymers moulding, etc.)

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Preparing and forming the system	**20**	**30**	**-**	**15**
PC1. investigate whether the robot and its peripheral equipment are responding to the programs' instructions	3	5	-	2
PC2. revise, repair or expand existing programs to increase operational efficiency or adapt to new requirements	3	5	-	2
PC3. repair or replace components as required	4	5	-	2
PC4. develop Human-Machine-Interface (HMI) applications for the users of the robot system, using HTML or other web technologies	5	5	-	4
PC5. advise on maintenance regimes to maximize efficiency and minimize disruption	5	10	-	5
NOS Total	**20**	**30**	**-**	**15**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6455
NOS Name	Preparation and forming
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

ASC/N6456: Finalize and deliver workpieces

Description

This NOS is about finalizing and delivering the parts.

Scope

The scope covers the following :

- Finalizing and delivering the parts

Elements and Performance Criteria

Finalizing and delivering the parts

To be competent, the user/individual on the job must be able to:

PC1. Clean parts

PC2. Deliver parts to the appropriate locations and/or personnel as required by the organization

PC3. Evaluate and report on factors and outcomes relevant to requirements and expectations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. The processes and procedures of post processing

KU2. The importance of completing work pieces to the required standard to the extent of their responsibilities

KU3. The circumstances in which referral should be made to other appropriate personnel

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Finalizing and delivering the parts	**20**	**30**	**-**	**10**
PC1. Clean parts	5	8	-	3
PC2. Deliver parts to the appropriate locations and/or personnel as required by the organization	7	10	-	3
PC3. Evaluate and report on factors and outcomes relevant to requirements and expectations	8	12	-	4
NOS Total	**20**	**30**	**-**	**10**

National Occupational Standards (NOS) Parameters

NOS Code	ASC/N6456
NOS Name	Finalize and deliver workpieces
Sector	Automotive
Sub-Sector	Manufacturing
Occupation	Production Engineering
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2026
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ASC/N9821.Work organization and management (Additive Manufacturing)	15	25	-	10	50	5
ASC/N9822.Communication and interpersonal skills (Additive Manufacturing)	20	30	-	10	60	5
ASC/N6457.Three D digitizing	30	50	-	20	100	15
ASC/N6453.Component optimization/structural optimization	20	30	-	10	60	25
ASC/N6454.Transfer-to-CAD and optimization	30	50	-	20	100	20
ASC/N6455.Preparation and forming	20	30	-	15	65	25
ASC/N6456.Finalize and deliver workpieces	20	30	-	10	60	5
Total	**155**	**245**	**-**	**95**	**495**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.